

Two Clustering Algorithms – K-Means and DBSCAN

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Background Information

Clustering is grouping together statistically similar data to highlight commonalities in various groups as a part of data analysis. Commonly used in unsupervised learning, this technique can find commonality within a dataset and group together datapoints without a target attribute. The sample dataset used for this project is the Kaggle.com Mall Customer dataset which consists of 200 customers' gender, age, annual income, and their spending score (from 1-100). The goal of this project is to find spending patterns based on customers' age, gender, and income, and develop actionable business recommendations for various groups of customers.

Data Exploration and Pre-processing

The data used was very clean and needed only minimal pre-processing. The only non-numerical attribute was the Gender column, which even as a categorical attribute is only binary. Thus, to help prevent this binary value from skewing the data, it was dropped from processing, however, it was later re-added back to provide a little more information with the visualization. The remaining data was all used to create a 3-D vector for each entry, containing the age, income, and spending score. This data was then normalized using Z-Score normalization to bring the values all within the range of -1 to 1, providing a clean cubic region of data, preventing the larger-scale attributes from also skewing the values. The Z-score formula used to normalize the age, income and spending column is $Z = \frac{(X-\mu)}{\sigma}$, where X is the original value from the dataset, μ is the mean of the dataset, and σ is the standard deviation.

After normalizing the data, it was run through different algorithms to determine the best parameters for each clustering algorithm used. For K-Means, a simple elbow plot was used to determine the best number of clusters, the K value, based on inertia values of each value. This

determined that the best number of clusters was 5, producing relatively clean clusters (*fig. 1*). For DBSCAN, first the data was run through a similar algorithm, a Knee plot, to determine the starting value for epsilon, and then through a quick sweep of similar values, both higher and lower, and combined with a variety of values for minimum samples. This determined that the best combination for both minimal noise and strong clusters was at an epsilon of approximately 0.634 and 6 minimum samples. (*fig. 2*)

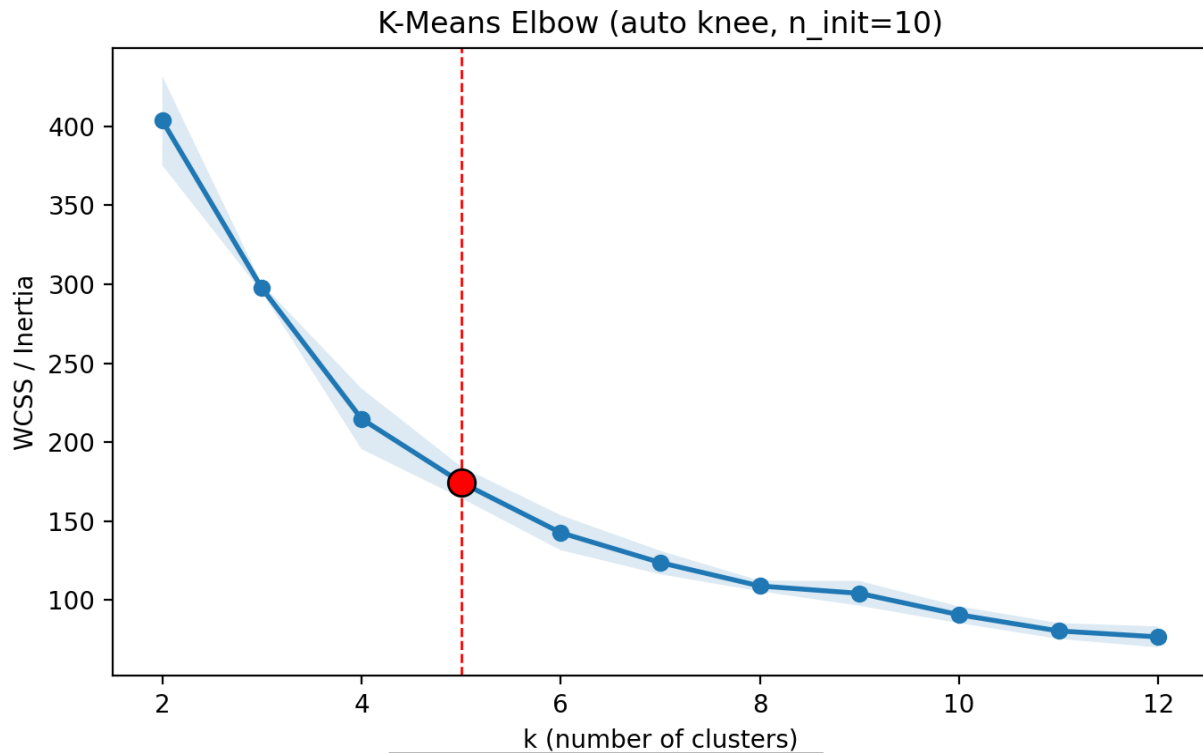


Figure 1. K-Means Elbow Plot

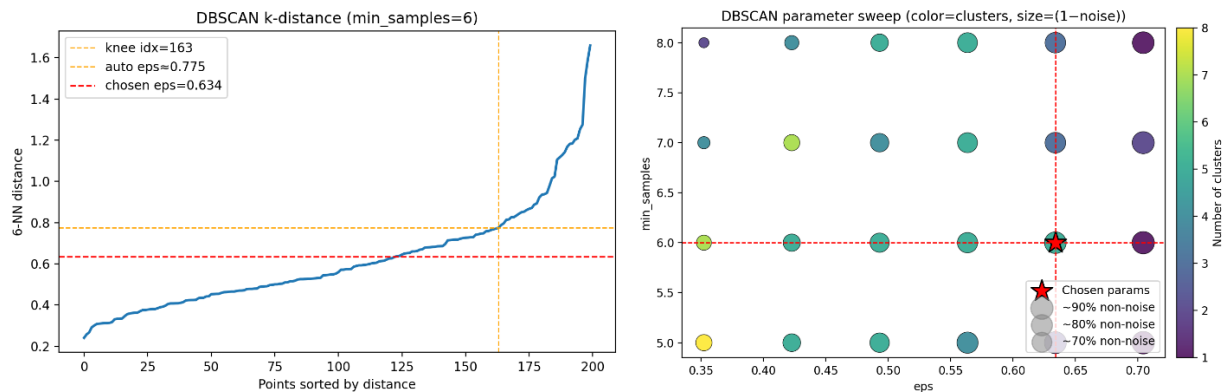


Figure 2. DBSCAN Knee Plot and Parameter Sweep

Methodologies

The data was processed using two different clustering algorithms, K-Means and DBSCAN (Density-Based Spatial Clustering of Applications with Noise). K-Means clustering partitions the dataset into k groups by iteratively refining centroid positions. The algorithm begins by selecting initial centroid locations (often using the k-means++ strategy to spread them out). Each data point is then assigned to the nearest centroid, forming clusters. The centroids are recalculated as the mean of the points assigned to them, and assignments are updated accordingly. This process repeats until convergence, defined as either reaching a maximum number of iterations or when centroids and labels stabilize. While initialization contains randomness, running multiple iterations with different seeds typically ensures stable, consistent results on well-separated data.

DBSCAN groups points based on density rather than a predefined number of clusters. It relies on two parameters: ϵ (epsilon), the maximum neighborhood radius, and *min_samples*, the minimum number of neighbors required to form a dense region. A point with at least *min_samples* neighbors within distance ϵ is considered a core point. Core points expand clusters by recursively connecting to neighboring points; points within ϵ of a cluster but lacking sufficient neighbors become border points. Points not belonging to any cluster are labeled as noise. Unlike K-Means, DBSCAN does not require the number of clusters in advance and is deterministic for fixed parameters and data.

Results

Both methods used created good, strong clusters, and had similar boundaries between clusters, identifying 5 core groups: young people who have low income, but spend a lot, people

of all ages who have low income and do not spend much at malls, a group of young people who have high incomes that spend a lot, a group across all ages who have median incomes between \$40,000 and \$70,000 who all have middle spending scores, between roughly 40 and 60, then a wealthy group of all ages who do not spend much. In addition to these groups showing up in both K-Means and DBSCAN, DBSCAN additionally identified 30 noise points, or 7.5% of the data, who are more outliers that do not really cluster into any groups. Most of these come from the wealthy low spending bracket, which tends to be much more spread out, however, there are some in all four of the corner groups.

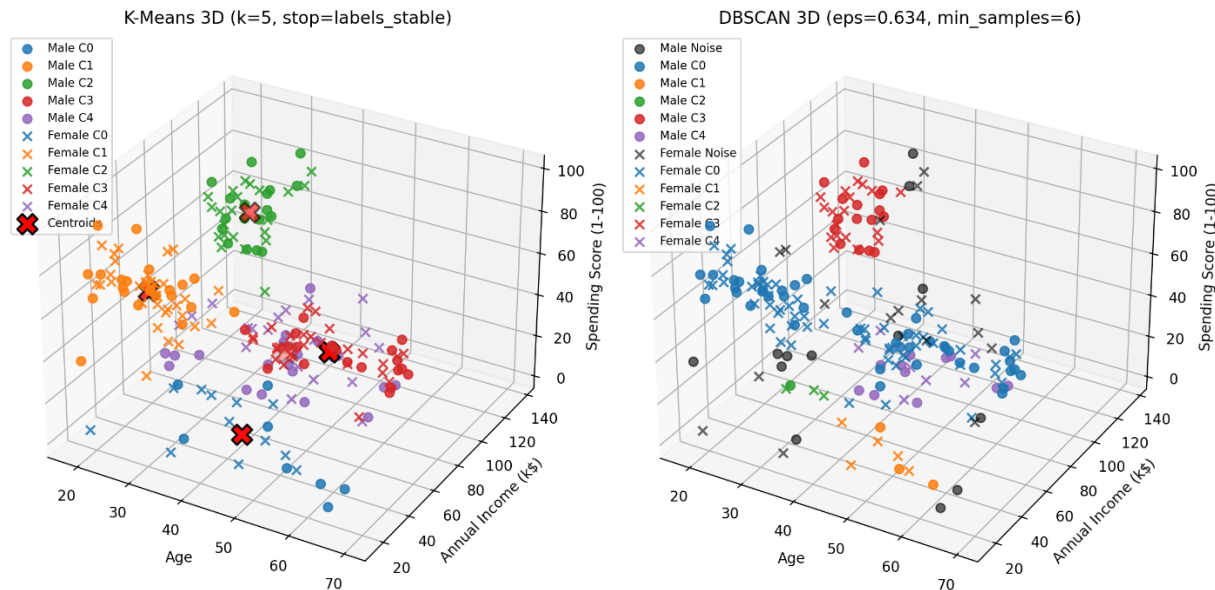


Figure 3: 3-D scatter plots showing clustering for both K-Means and DBSCAN

To evaluate the resulting clusters, the K-Means was tested with the silhouette score. This method takes an average across all points based on each point's distance from all other points in its cluster, its intra-cluster distance, versus the distance from all data points in the nearest neighboring cluster, the inter-cluster distance. These scores range from -1 to 1, with negative numbers usually indicating some level of incorrect assignment, and higher positive values indicating good clustering. The silhouette score was roughly 0.42, which is a moderate score,

showing decent cohesion within clusters and fair separation between clusters, providing actionable information.

DBSCAN, however, does not typically score well with silhouette, due to the method used for determining clusters, and the addition of noise, since silhouette does not take such noise data into account. Therefore, DBSCAN was evaluated using two index values, the Davies-Bouldin Index (DBI), and the Calinski-Harabasz Index (CHI). DBI is like silhouette scoring, however, instead of basing it on point to point, it uses point to centroid distance. The formula for DBI is

$$\frac{1}{n} \sum_{i=1}^n \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d(c_i, c_j)} \right)$$

shown in figure 4, with σ_i is the average distance of all elements in cluster 'i' to centroid c_i , and ' $d(c_i, c_j)$ ' is the distance between

Figure 4: DBI formula

centroids c_i and c_j . Thus, lower scores are better, and for this data

and clustering, the DBI was approximately 1.12, which indicates that the clusters are well separated in respect to the intra-cluster distances.

The CHI is the normalized ration of between-cluster separation to the within-cluster separation, and is given by the formula in figure 5, with 'k' the number of clusters, and 'n' the number of data points. The BCSS and WCSS are the between-cluster and within-cluster sum of squares,

$$CH = \frac{BCSS/(k-1)}{WCSS/(n-k)}$$

Figure 5: CHI formula

respectively, and represent the weighted sum of squared Euclidian distances. For between-cluster, it is between each cluster centroid and the data centroid, weighed for the number of points in the cluster, meaning higher numbers are better. Within-cluster is between each data point and the cluster centroid; thus, lower numbers are stronger. With these two values, overall, a larger number indicates stronger clusters and can increase without bound. The DBSCAN results had a CHI of approximately 34.3, which, given the size of dataset, shows a strong structure evident in the clustering.

Business Insights

Based on this dataset, we have found that the younger generation has the highest spending score out of all age groups, regardless of their income. In the sense of statistics, this group are often seen as an outlier group. However, in the context of finding the spending habit within a group of consumers, this outlier can benefit the businesses. As a result of studying the spending habits of this population, businesses can plan a variety of selling strategies to attract more young customers, such as discounts and bundle deals to catch their attention. Furthermore, it is within reasonable assumption that the younger generation will spend money if they find the product to be good-looking and functioning solely by looking at the packaging. To businesses manufacturing consumer products, focusing on designing a functioning and fashionable product will be able to catch the majority of the younger generation's attention and sparks their desire to spend money on that certain product.

As far as the other two income groups, people who make between about \$40k to \$70k tend to be more conservative on spending. This group of consumers tends to only make purchases that they deemed to be necessary to their daily life. Manufacturers and businesses focusing on this group of consumers should pay closer attention to the price of their products, all while maintaining the quality of their products every customer is looking forward to. Unlike the younger generation, manufacturers and businesses selling to this group of consumers should focus on both the price and quality of their products while still maintaining a decent profit margin.

Conclusion

Applying clustering techniques in this customer database is an effective way to find commonality in both the qualitative and quantitative traits in a database. In the machine learning world, these two techniques can increase efficiency when training a machine learning model as it can group and label similar datapoints without programmers having to feed in these labeled data that requires hours of manual data sorting and labeling. Overall, these two clustering techniques can greatly reduce human intervention in designing and training a machine learning model.